



The emergence of entrepreneurial ecosystems: A complex adaptive systems approach

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ABSTRACT

Entrepreneurial ecosystems are receiving heightened attention from scholars and practitioners. Studies have focused on isolating entrepreneurial ecosystems' components; however, prior research has not offered a theory of entrepreneurial ecosystems that embraces their complexity. To address this omission in ecosystems research, we contend that entrepreneurial ecosystems can be more fully understood if examined through the lens of complexity science and conceptualized as complex adaptive systems. We contribute to entrepreneurship research by developing a complexity-based definition of entrepreneurial ecosystems. Building on this definition, we connect the research on entrepreneur- and venture-level complexity to work on entrepreneurial ecosystems and propose three related forces that will influence entrepreneurial ecosystem emergence: intentionality of entrepreneurs, coherence of entrepreneurial activities, and injections of resources. Beyond developing theory, we describe how scholars can examine entrepreneurial ecosystems as complex systems using qualitative comparative analysis, agent-based modeling, and interpretivist methods. Our theorizing also has implications for entrepreneurs and policymakers.

1. Introduction

The linkages between entrepreneurial activity – the creation of innovative organizations, products, and initiatives that generate value for society – and regional economic development are receiving growing interest from scholars and policymakers (Audretsch, Keilbach, & Lehmann, 2006; Galindo & Méndez, 2014). Acknowledging the social and cultural embeddedness of entrepreneurial activities, researchers have increasingly shifted their focus from studies of entrepreneurs and ventures to the creation of entrepreneurial ecosystems (EEs): the sets of actors, institutions, social networks, and cultural values that produce and sustain entrepreneurial activity (e.g., Ács, Stam, Audretsch, & O'Connor, 2017; Auerswald, 2015; Brown & Mason, 2017; Stam, 2015). The EE approach draws attention to the configuration of individual, organizational, and societal forces that is necessary to promote and support entrepreneurial activities (Berger & Kuckertz, 2016a; Roundy, Brockman, & Bradshaw, 2017; Spiegel, 2017; Spilling, 1996).

In studies of EEs, scholars have focused primarily on identifying the core attributes of established ecosystems, such as Tel Aviv (Klingler-Vidra, Kenney, & Breznitz, 2016) and Edinburgh (Spiegel, 2016). For

instance, Bahrami and Evans (1995) found that the Silicon Valley ecosystem's critical features include a deep reservoir of venture capital, knowledgeable labor, research institutions, professional services infrastructure, and lead users of innovations (also cf. Kenney & Von Burg, 1999). Although these studies and others (e.g., Isenberg, 2010) have isolated the key components of several high-profile ecosystems, it is increasingly clear that to understand EEs and how they emerge, it is necessary to go beyond producing lists of attributes (Auerswald, 2015). Indeed, Mack and Mayer (2016: 2118) argue that a key limitation of the current work on EEs is its focus on “documenting the presence of system components, [with] little understanding of interdependencies between components.”

We agree with these scholars and others (e.g., Spiegel, 2017; Stam, 2015) that what is missing from the prior work on EEs is a guiding theoretical framework that acknowledges the complexity of the phenomenon. EEs have been the target of academic and practitioner attention for over 25 years; however, by not focusing on the complex interactions among agents, organizations, and socio-cultural forces, we know surprisingly little about how ecosystems emerge (Auerswald, 2015). Attempts have been made to explain entrepreneur- and venture-

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level emergence (e.g., Lichtenstein, Carter, Dooley, & Gartner, 2007; McKelvey, 2004); however, a theory has not been put forth that specifically addresses the complexity and emergence of EEs.

A clue as to the type of theory that could illuminate the study of EEs and provide insights into their emergence can be found in the aforementioned calls for studying the complex constellation of connections among ecosystem components, which suggest that EEs are best treated as *systems* and that systems theory, an analytical approach representing phenomena as sets of stocks and flows regulated by interactions (e.g., Hartvigsen, Kinzig, & Peterson, 1998), might provide an appropriate lens for understanding EEs. Work in systems theory has taken two approaches. The first approach assumes that systems are commonly in (or near) equilibrium, which negates the need to examine dynamic relationships and nonlinear interactions among the systems' elements and instead focuses on isolating and parameterizing stable, individual components (Manson, 2001). Although the “simple” systems approach is appropriate for explaining the behavior of some types of systems, a second approach used by a variety of disciplines, including biology, ecology, chemistry, economics, and management (cf. Eisenhardt & Piezunka, 2011), suggests that there is a second type of system that does not operate at equilibrium. There is a subset of these non-equilibrium systems – *complex adaptive systems* (CAS) – that cannot be explained using general systems theory.

The study of complex adaptive systems – systems in which macro-level behaviors both emerge from and influence the micro-level interactions of the elements of the system (Levin, 2002; Lissack & Letiche, 2002) – has led to an interdisciplinary branch of scholarship referred to as *complexity science* (Manson, 2001). The aim of this research is to provide a framework for analyzing the characteristics of complex systems, such as nonlinearity, self-organization, cross-scale interactions, and emergence (Arthur, 1999; Berger & Kuckertz, 2016b).

We theorize that EEs are complex adaptive systems. Moreover, we contend that by analyzing EEs through the lens of complexity science we can move past lists of ecosystem components, provide a framework that can be used to respond to calls for studies of the emergence of EEs, and connect micro- and macro-level research in entrepreneurship. Thus, in this paper, we build on complexity science to offer a framework for the study of EEs.

The remainder of the paper is structured as follows. First, we review work on EEs, highlighting the approach's history and key insights. We then review research on complexity and entrepreneurship focused on the individual- and venture-levels, describe how this research has not been extended to the ecosystem-level, and explain how complexity science can be used to connect work on the emergence of innovations, entrepreneurs, and ventures to research on the emergence of EEs. Next, we apply the lens of complexity to EEs, establish the appropriateness of conceptualizing EEs as complex adaptive systems, and introduce a complexity-based definition of EEs. We then theorize about the emergence of EEs as complex systems. Finally, we discuss the contributions of our theorizing for scholars, propose three methods for studying EEs as complex systems, and suggest implications for practitioners and policymakers.

2. Literature review

2.1. Entrepreneurial ecosystems

Bahrami and Evans (1995) were the first in the academic entrepreneurship literature to invoke the term “ecosystem” in their study of Silicon Valley. Similarly, Spilling (1996: 91) emphasized the “entrepreneurial system,” describing it as the actors, roles, and environmental factors that interact to determine the entrepreneurial performance of a region. Motoyama and Knowlton (2017) focused more directly on the connections among the agents in an EE. In doing so, they point out that while the system of connections among the agents in an EE is important, very little investigation into the complexities of this

system has been done. They also note that the emergence of EEs has been recognized by researchers; however, direct theorizing about the emergence process (i.e., how an ecosystem form) is lacking (Motoyama & Knowlton, 2017).

Several themes appear in prior studies of EEs. Findings suggest that EEs emerge over time through multiple components and micro-level processes (e.g., the intentions of entrepreneurs), meso-level processes (e.g., the provisioning of resources to entrepreneurs from support organizations), and macro-level processes (e.g., the influence of ecosystem culture) (e.g., Isenberg, 2010). There is also some recognition of the complexity of the agents in an EE and that the interactions among them are paramount in developing the ecosystem (Motoyama & Knowlton, 2017; Spilling, 1996). Finally, although there is a vibrant stream of research on EEs, scholars have yet to converge on a widely accepted definition and one that is theoretically grounded. In sum, researchers have produced insights regarding the general nature of EEs, including identifying their most common components and drawing attention to the relevance of the relationships among agents. To move forward, however, what is needed is a theoretical framework that ties together disparate insights. In the next section, we suggest that complex adaptive systems theory provides such a framework.

2.2. Complexity science, entrepreneurship, and emergence

A common conceptual lens – complexity science – has developed to analyze systems in which the interactions between components result in the emergence of novel, seemingly unpredictable patterns, behaviors, and structures (Anderson, Drakopoulou Dodd, & Jack, 2012; Fuller & Moran, 2001). In such systems, the patterns of action produced at one level both emerge from and are influenced by processes operating at different levels and by the behaviors of the overall system (Hartvigsen et al., 1998; Lissack & Letiche, 2002), a characteristic referred to as *complexity* (Arthur, 1999; Lansing, 2003). Systems that exhibit complexity and are adaptive (i.e., have the capacity to change based on experience) are referred to as complex adaptive systems (Schindehutte & Morris, 2009). In such systems, the individual components are constantly reacting to one another (and to the environment) across levels, modifying the system and its response to disturbances and allowing it to adapt to changes (Messier & Puettmann, 2011: 250).

At the core of both complexity science and entrepreneurship scholarship is a focus on the concept of emergence, “the creation of new ‘order’ – structures, processes, and system-wide properties that come into being within and across system levels” (Lichtenstein, 2011a: 486). Emergence is central to entrepreneurship research, which has emphasized “the coming-into-being of new organizational means (e.g., resources) that in turn lead to the creation of new entities, e.g., technologies, firms, networks, clusters and markets, industries, [and] institutions” (Gartner, 1993; Lichtenstein, 2011a: 474). Studies of complexity science and entrepreneurship also align in their emphasis on how innovations both influence and are an outcome of emergence (Fleming & Sorenson, 2001; Garud & Karnøe, 2003; McKelvey, 2004). Because of the conceptual fit between complexity and entrepreneurship, complexity science has been used, primarily at the individual- and organizational-levels, to study the emergence of entrepreneurial behaviors and new ventures (Lichtenstein, 2011b; Lichtenstein et al., 2007).

The application of complexity science to the study of entrepreneurs and new ventures was, in part, a reaction to the focus in most entrepreneurship research on specific components of the entrepreneurship process (e.g., developing a business model, hiring early-stage employees, attracting investment) instead of on the recursive and nonlinear interactions among these sets of activities (Gartner & Carter, 2003). Thus, before the complexity lens was used, studies of founder- and organization-level entrepreneurship had a similar focus as prior EE research: identifying the components of the entrepreneurial process rather than exploring “interactions and emergent phenomena at multiple levels of analysis” and highlighting the importance of “nonlinear

and unpredictable processes that generate emergent order” (Lichtenstein, 2011a: 473).

In the section that follows, we build on the developments in complexity science to examine the appropriateness of conceptualizing entrepreneurial ecosystems as complex adaptive systems. We use these arguments to motivate and introduce a complexity-based definition of entrepreneurial ecosystems, which we then build upon to theorize about the emergence of EEs.

3. Entrepreneurial ecosystems as complex adaptive systems

Diverse phenomena can be classified as complex adaptive systems (CAS). Despite differences in their structures, scales, and agents, CAS share six properties: *self-organization*, *open-but-distinct boundaries*, *complex components*, *nonlinearity*, *adaptability*, and *sensitivity to initial conditions*. In the sections that follow, we propose that EEs possess these properties and, thus, can be conceptualized as complex adaptive systems.

3.1. Emergence through “self-” organization

Entrepreneurial ecosystems are not governed by a global controller, a single leader, or an organization (Isenberg, 2010). As we explain, the order that emerges in EEs does so, in large part, from the uncoordinated, semi-autonomous actions of individual agents. Order is not imposed by a system “manager.” A high-profile entrepreneur, investor, or philanthropist might play a large role in an ecosystem in terms of capital, legitimacy, or connections (Feldman & Zoller, 2012); however, no single agent can be said to be in control of the behaviors of the EE or its actors. Similarly, although some organizations (e.g., an incubator) might be more influential than others (Albort-Morant & Oghazi, 2016), there is generally not a single entity that universally directs the EE's activities, which is important because it implies that the behaviors and structure of the system are emergent and arise from self-organization rather than “top-down” control (Nicolis & Prigogine, 1977). As we describe in the next section, an EE emerges from the micro-interactions of its individual participants, which, when aggregated, construct the complex system. Indeed, anecdotal evidence suggests that if one agent or organization is too heavy-handed in its attempts to direct an EE, it can harm its cohesiveness and functioning (Feld, 2012).

Proposition 1. Entrepreneurial ecosystems exhibit self-organization: order that emerges does so without a global controller.

3.2. Open-but-distinct boundaries

Examining EEs as complex adaptive systems can help to address whether there are boundaries to such systems and, if so, how to identify them. In general, there is a critical distinction between closed, simple systems and complex adaptive systems (Cilliers, 1998). Simple systems often have defined boundaries that clearly separate what is and is not a system element. In contrast, the essence of complex systems is that they are open, do not follow the predictable entropic path of closed systems and are “far from equilibrium” (Fuller & Moran, 2001; Prigogine & Stengers, 1985). Although the borders of EEs, as complex adaptive systems, are ill-defined, borders *do* exist as both geographic and socio-cultural entities.

An early conceptualization of an EE described it as “an *inter-connected* group of actors in a local geographic community” (Cohen, 2006: 3; emphasis added). The geographical boundary of an EE is emphasized in several other studies and in practitioner work (Feld, 2012; Johannisson, 2000; Mack & Mayer, 2016; Spigel, 2017). The socio-cultural component of an EE's boundaries has received less attention. Alvedalen and Boschma (2017: 891) describe entrepreneurship as “embedded in social relationships – a network”; however, as they

also note, the EE literature has not yet produced a comprehensive networks approach. Nevertheless, EE boundaries may be defined, in part, by network connections with external agents and systems. These connection-type boundaries constitute the interface through which exogenous resources and information are brought into the EE.

Furthermore, as open systems, EEs are dependent on resource providers outside the system, and influenced by resource flows into the ecosystem. As described later in this section, a characteristic of EEs is that there are feedback loops among the system elements and among EEs and resource providers (e.g., between the density of entrepreneurs and the amount of available investment). There is also feedback between the ecosystem and the elements *outside* it, which helps define its boundaries. For instance, there are feedback loops among the dynamics attracting financial and human capital to an EE and regulating the supply of capital and talent into and out of the system (e.g., resources entering an ecosystem may attract more resources; however, too many resources can impede the flow of new resources), further illustrating the function that boundaries serve. Indeed, one of the reasons for establishing and promoting an EE is that doing so helps attract resources to the region in which the ecosystem is embedded (e.g., Mason & Brown, 2014). This implies a boundary, one that resources must “cross” to become part of the EE and flow into and out of it.

The “in” and “out” language used to describe this process is partially related to geography; however, it is also associated with other dimensions. Specifically, we assert that the boundaries of EEs are multi-dimensional and dependent on several factors, such as the extent to which agents share certain socio-cultural characteristics with other agents, such as guiding rule sets, logics, and values (cf. Roundy, 2017). If agents have these characteristics, it will influence the degree to which their intentions and behaviors have coherence with other agents in the system (Lissack & Letiche, 2002). For instance, many EEs have a physical geographic epicenter, such as an explicitly defined “innovation district” (cf. Cohen & Munoz, 2016). Agents located near this epicenter are more likely to engage in activities correlated with one another (e.g., founding businesses based on the technologies emphasized in the innovation district). However, as agents become increasingly geographically distant from this collection of highly cohered agents, their relationship with the activities of the EE will weaken.

Similarly, socio-cultural characteristics, like shared values, will also influence membership within an EE. For instance, the Boulder, Colorado, ecosystem is described as “defined by a strong sense of collaboration and implicit rules such as ‘giving before you get.’ If you contribute, you are rewarded, often in unexpected ways. At the same time [...] if you aren't sincere, constructive, and collaborative, the community behaves accordingly” (Feld, 2012: 7). Thus, agents not demonstrating certain values will be considered “outside” an ecosystem. Overall, as an agent gets further from either the geographic epicenter of an EE (often synonymous with a high density of entrepreneurs; Tödtling & Wanzelböck, 2003) and/or is less aligned with its social-cultural values, narratives, or dominant rule set, the effects of the ecosystem on the agent will be weaker; in contrast, the more aligned an agent is with the dominant forces shaping the EE, the greater the effect of the EE on the agent.

Proposition 2. Entrepreneurial ecosystems exhibit open-but-distinct boundaries based on geographic and socio-cultural characteristics.

3.3. Complex components

Examining the components of an EE is the most direct connection between complexity science and prior work on EEs (e.g., Isenberg, 2010). One component of an EE is the agents who comprise it. An ecosystem's agents, including entrepreneurs, investors, mentors, and other resource providers, are heterogeneous in their attributes (e.g., opportunity recognition abilities and resource endowments; Roundy et al., *in press*), their interactions with the external environment (e.g.,

connections to resource providers outside the ecosystem), and their relationships with other agents (e.g., the breadth of their intra-EE social networks; Chen, Chang, & Lee, 2015). Despite the heterogeneity among the individual agents in an EE, there is enough homogeneity in agent-types to assign agents to categories based on similarities in intentions, behaviors, and activities. For instance, although there is variation among entrepreneurs in their aims, abilities, and types of ventures (Roundy et al., *in press*), individuals performing these roles share fundamental similarities that allow them to be assigned to a category, “entrepreneur,” which distinguishes them from other groups, such as investors. In a complex adaptive system, agents are not role-exclusive and may have multiple roles (e.g., an entrepreneur who is also an angel investor); however, categories of like-roles can be identified.

Beyond heterogeneity, the components of an EE operate at different levels, including the individual- (e.g., founder), organization (e.g., accelerator), and para-organization (e.g., Chamber of Commerce), which aligns with complex systems being driven by multi-level activity (Lichtenstein, 2011b). The features of an EE will operate differently depending on multiple size- and time-scales. For example, knowledge diffusion, learning, innovation transfer, and the effect of culture can change as the system grows (e.g., Bottazzi & Peri, 2003). Similarly, technological innovations can develop at the scale of months or years; however, they can shape entrepreneurial decisions that occur on the scale of days or weeks (e.g., Attewell, 1992). Some EE agents' activities occur at a rapid tempo (e.g., entrepreneurs' testing of business model hypotheses), while other actions occur over longer periods, such as the self-reinforcing cycle by which founders become mentors (Wolfe, 2002). These components, often complex systems in themselves, may be conceptualized as aggregated systems, which coalesce through the system-wide action of feedback loops (Byrne & Callaghan, 2013). Overall, the components of EEs possess the same properties as the components of other complex adaptive systems.

Proposition 3. Entrepreneurial ecosystems exhibit complexity in their components.

3.4. Nonlinear dynamics

EEs exhibit another property of complex adaptive systems: the interdependent components of EEs lead to nonlinear dynamics and feedback loops, which occur when an activity feeds back on itself either directly or after intervening processes (Cilliers, 1998). Indeed, it is out of large-scale, nonlinear interactions that complexity emerges; the nonproportionality of such systems produces situations in which small inputs or forces can have large impacts (and vice versa) (Berger & Kuckertz, 2016b; Lichtenstein, 2000). Specifically, a *positive* feedback loop is enhancing and results in system behaviors increasing or decreasing indefinitely (Cilliers, 1998; Manson, 2001). Positive feedback loops are described as *autocatalytic* and represent the ability of a system to grow and be stimulated from within (Morrison, 2008). The number of entrepreneurs in an EE, for instance, is associated with an increase in the number of investors (as deal flow increases and as some entrepreneurs exit their ventures and assume the role of investor); increases in the number of investors, and the amount of capital available, will attract more entrepreneurs to the region, which in turn will attract more investment (cf. Lipper & Sommer, 2002). Thus, the relationship between the number of entrepreneurs and investors is self-reinforcing, at least at the system level. A similar, self-reinforcing feedback loop has been argued to exist between the growth in the number of entrepreneurs in an EE and an ecosystem's arts and entertainment venues (Phillips, 2011).

In complex systems, positive feedback processes can also lead to the emergence of new activities, which then lead to an increase in other activities (Lichtenstein et al., 2007). For instance, continuing with the above example, as the number of entrepreneurs in an EE increases, this can lead to more early-stage investors; and the presence of more

investors can cause new types of entrepreneurial activity to emerge. Similarly, as entrepreneurs in an EE find success, some proportion may want to “give back” (e.g., Feld, 2012) to the community by helping entrepreneurs as mentors; some of these entrepreneurs, in turn, find success and themselves become mentors – thus, leading to the self-reinforcing growth of the mentor network.

In contrast, a negative feedback loop between two components causes one component to move towards an equilibrium or steady state, at least temporarily, rather than triggering an indefinite increase or decrease (Lichtenstein & Plowman, 2009). For example, local housing that is available and affordable can be necessary for the growth of a talented worker pool. If such housing exists, then it is possible for this pool of workers to contribute to the growth of an EE. However, without the availability of affordable housing, the worker pool will reach a point of stability whereby growth ceases, but a decrease is not imminent. Thus, a negative feedback loop exists. In sum, these arguments suggest the following:

Proposition 4. Entrepreneurial ecosystems exhibit nonlinearity in the relationships among ecosystem components.

3.5. Adaptability through dynamic interactions

In addition to the complexity that arises from the nonlinear interactions of a system's components, such interactions can also produce adaptability (Cilliers, 1998; McKelvey, 2004). Through interactions with one another, the actions of agents within an EE will produce continuous modifications to the system, which shape how the system responds to endogenous and exogenous disturbances and allow it to adapt to changing and novel conditions (Messier & Puettmann, 2011). Specifically, as the entrepreneurs, resource providers, policy makers, and other agents in an EE respond to disturbances or to injections of resources into the system, the network that connects the agents can itself morph and evolve. These changes ascend through the levels of the EE and create the means by which the system can adapt to changing circumstances. Thus, system-level adaptability emerges from behaviors at lower levels, even as the agents comprising those levels are themselves influenced by system-level changes. The “interpenetration of levels” (Seo & Creed, 2002: 222), with the consequent feedback effects among them, establishes nonlinear relationships among agents (Byrne & Callaghan, 2013). For example, one function of the mentors in an EE is strengthening the density of the system's entrepreneurial network and, specifically, expanding entrepreneurs' connections to resource providers (both within and outside the EE) (Motoyama, Fetsch, Jackson, & Wiens, 2016). Doing so increases entrepreneurs' range of actions, thereby improving their flexibility and, in turn, influencing the overall adaptability of the system.

Another example is that entrepreneurs may realize that their EE lacks a type of human capital (e.g., software development abilities), which may initially limit their ability to found certain types of ventures (e.g., high-technology firms). As entrepreneurs connect with other agents, they will have an opportunity to communicate about this gap in the ecosystem's human capital. A growing number of agents communicating about the topic will garner increasing attention to it. At some point, organizational actors (e.g., the Chamber of Commerce) may begin to address the problem (e.g., by offering incentives to attract individuals from outside the EE with the requisite human capital or implementing programs that develop talent within the system). Eventually, if the initiatives are successful, the stock of this type of human capital will increase, which represents a change in the ecosystem's agents. This change will, in turn, increase the range of possible behaviors in the EE and make adaptation possible. Continuing with the previous example, before the influx of new human capital, a simple rule may pervade the system and influence agents' actions: “when founding a business in this ecosystem, creating a cutting-edge, technology company is not feasible.” If this rule is changed, it will have a ripple-effect

throughout the system that results in adaptation. Adaptability is thus a product of the micro- and macro-level interactions in the EE: the actions of individual agents both help to create system-level rules and, in turn, are influenced by such rules.

Proposition 5. Entrepreneurial ecosystems exhibit adaptability: the actions of individual agents produce continuous modifications to the ecosystem that allow it to adapt to changing conditions.

3.6. Sensitivity to initial conditions

The nonlinear dynamics of CAS are accompanied by a final property that is also shared by EEs: sensitivity to starting conditions (McKelvey, 2004). Small changes in the initial configuration of an EE can, over time, have large and unexpected effects on its later development (e.g., Gray, Golob, & Markusen, 1996). These dynamics can lead to path dependency, when a system's behavioral responses become locked-in to a narrow trajectory because of historical experiences (Kenney & Von Burg, 1999). With path dependence, accidents of history become embedded in the system, constrain its future behaviors, and cause its development to diverge from systems that do not share the same experiences (Levin, 1998). For instance, the idiosyncratic configuration of circumstances that helped the Silicon Valley ecosystem coalesce (e.g., the decisions of several semiconductor companies to locate in the Santa Clara Valley in the 1950s; Bahrami & Evans, 1995), had a profound impact on shaping the development of the ecosystem. Silicon Valley's heavy (parallel) emphasis on high-technology ventures and venture capital, also a positive feedback loop, has led to an EE capable of producing an unprecedented number of innovations. However, the development of the ecosystem has also been associated with a concomitant increase in the region's cost of living, which may act as a restraint on other activities (cf. Neck, Meyer, Cohen, & Corbett, 2004).

As Prigogine and Stengers (1997) cautioned scholars, there is an “arrow in time” involved in many system processes, which is fundamental to understanding not only the processes' complexity but also how the system has achieved its current state. For such processes, there is no reversibility; thus, the historical origins of events and decisions are relevant. For instance, because of the locking-in effect of previous circumstances, it can be difficult for an EE that has come to focus on one form of entrepreneurship or type of technology to change course and focus on an entirely different type. As with an organization, the past provides the foundation upon which an EE is built and, thus, is an important source of its identity, reputation, and culture (Walsh & Ungson, 1991); however, the past can also represent a constraint on the future focus and adaptability of the EE. Decisions made during the formative years of the EE can be irreversible. For example, the creation of a technology accelerator focusing on an industry cannot be undone. Although the accelerator can be closed, the time and resources invested in its creation and the attention it commands from ecosystem participants influence the evolution of the EE in ways that are not reversible. As a more specific example, Seattle, Washington, became a vibrant EE “on Boeing's wide shoulders” as the city's software hub grew out of the company's initial, internal need for software, which eventually led to the specialized pool of workers and infrastructure needed for the technology sector to develop (Gray et al., 1996: 662). In contrast, Detroit, Michigan, is experiencing the negative influence of the “past dominance of the ‘Big Three’ automobile companies,” which, rather than promoting future EE development, “suppressed diversified growth and where [...] strategies [adopted] in the profit-squeeze era initiated a downward cycle” in the region (Gray et al., 1996). Thus, the same initial condition (in this case, the location of large firms) can lead to very different outcomes.

Proposition 6. Entrepreneurial ecosystems exhibit sensitivity to initial conditions.

Building on the above arguments about the properties of EEs as

complex adaptive systems and Propositions 1–6, we propose the following complexity-based definition:

An entrepreneurial ecosystem is a self-organized, adaptive, and geographically bounded community of complex agents operating at multiple, aggregated levels, whose non-linear interactions result in the patterns of activities through which new ventures form and dissolve over time.

Using this definition, in the next section we build on complexity research at the founder and venture (i.e., “micro”) levels and connect these dynamics to ecosystem (i.e., “macro”) level processes. We argue that the interplay between these dynamics leads to the emergence of EEs.

4. The emergence of entrepreneurial ecosystems

In the previous section, we sought to establish that entrepreneurial ecosystems are appropriately conceptualized as complex adaptive systems. Beyond merely an intellectual exercise, this is a necessary step in explaining the emergence of EEs. Emergence, the “process by which patterns or global-level structures arise from interactive local level processes” (Mihata, 1997: 31), is at the core of both entrepreneurship and complexity science (Lichtenstein, 2011b). Entrepreneurship-oriented emergence research has focused on the individual- and venture-levels. For example, scholars have examined how novel business ideas, practices, and ventures emerge from the micro-level cognitions and behaviors of entrepreneurs (e.g., Gartner & Carter, 2003; Lichtenstein, 2016; Steyaert, 2007). Emergence is also central to complexity science; indeed, complexity has been described as the study of emergent processes in complex systems (Lichtenstein, 2000). Since, as we have argued, EEs are appropriately conceptualized as complex adaptive systems, it is a natural – and critical – starting point to understand their emergence.

4.1. The micro-foundations of entrepreneurial ecosystem emergence

In theorizing about EE emergence in the sections that follow, we build on research on individual- and venture-level entrepreneurial emergence and on the role of intention and action in the creation of new ventures (e.g., Encinar & Muñoz, 2006; Lichtenstein, 2011a, 2011b). We propose that three related forces work in tandem to influence the emergence of EEs: the intentionality and adaptive tensions of entrepreneurs, the coherence of entrepreneurial activities, and injections of resources into the ecosystem. We theorize that the lens of complex adaptive systems can be used to understand the linkages between these forces that operate in EEs across levels.

4.1.1. Entrepreneur's intentionality and adaptive tensions

The first step towards understanding the emergence of a complex adaptive system is to identify the system's key agents, the forces influencing agents' cognition and behaviors, and the level at which these forces operate (cf. Bonabeau, 2002). Although there are numerous types of actors in EEs, we contend that entrepreneurs are the agents driving the creation of the complex system. It is entrepreneurs' intentionality, “the tendency towards a goal that first appears in the individual's mind as a purpose” (Krueger, Reilly, & Carsrud, 2000; Muñoz & Encinar, 2014: 323) that acts as a motivating force in an ecosystem and contributes to EE emergence.

Scholars adopting a complexity lens have argued that entrepreneurs create new ventures because of “adaptive tensions” – internal states of tension that are triggered by an external source and motivate a creative response by the entrepreneur (Lichtenstein et al., 2007; McKelvey, 2004). As Lichtenstein et al. (2007: 237) argue, “in the context of a nascent entrepreneur, adaptive tension is created through a perceived opportunity or by a personal aspiration to start a business.” Adaptive tensions are the result of “energy differentials,” which occur when there are discernable differences between the resources within entrepreneurs

and under their control, and new pools of resources that they desire to access in the pursuit of opportunities (Lichtenstein, 2011b: 4).

Adaptive tensions are critical to emergence because they are tied to entrepreneurial intentions. When entrepreneurs identify, create, and respond to market opportunities, they exhibit intentionality in formulating implicit and explicit action plans (i.e., projective linkages of actions to objectives; Encinar & Muñoz, 2006; Zapkau, Schwens, Steinmetz, & Kabst, 2015). The differential between the intentions of entrepreneurs and the opportunities on which they seek to capitalize generates pressures to act (Lichtenstein, 2011b). Specifically, based on intentions, entrepreneurs form plans as a means of searching for their business models (for example, through experimentation-based methods such as “Lean Startup”; Blank, 2013; Ries, 2011). When entrepreneurs deploy their plans, planned action becomes actual activity (Muñoz & Encinar, 2014). However, because of uncertainty, there is a discrepancy between planned and actual outcomes. As entrepreneurs take action, they compare expectations, based largely on initial intentions, to their outcomes. Surprises in this process are viewed as opportunities to leverage new information and can result in entrepreneurial learning – changing future actions based on the difference between *ex-ante* action plans and *ex-post* results (Minniti & Bygrave, 2001).

The entrepreneurial learning process, in conjunction with an entrepreneur's creativity, produces innovations, as the outcomes of an entrepreneur's plans are either reinforced by the evidence collected during trials or not (Osterwalder, 2004). Reinforcement occurs through a “successful” trial or through an ostensibly “failed” trial that finds unexpected success. Thus, mismatches between intentions and outcomes are critical because they not only provide the opportunity for learning (after which knowledge can be passed on to other agents in the system), but uncertain outcomes mean that the pursuit of opportunities will result in “errors,” which result in unexpected innovations being introduced into the system (Petkova, 2009).

4.2. The aggregation of individual entrepreneurial actions to entrepreneurial ecosystems

For an EE to emerge, however, is not simply a matter of individuals (or teams) creating ventures; entrepreneurs across ventures must also operate according to some degree of common behaviors, values, and methods (Lichtenstein et al., 2007; McKelvey, 1999). In other words, there is a degree of “coherence” – or correlation – across agents' activities, which gives shape to an ecosystem as an *interconnected system*. Indeed, despite the differences that exist across levels and scales, the agents that comprise EEs display coherence, which can be conceptualized as the degree of association between the components of a complex system that causes them to coalesce into a group (i.e., the system) rather than remain autonomous and independent (Manrubia, Mikhailov, & Zanette, 2004).

4.2.1. The coherence of entrepreneurial activities

The agents in an EE can gain coherence from several factors. First, building from the micro-level of the ecosystem, many agents will share a common set of intentions and action plans – founding and growing new organizations – which not only cause them to engage in similar behaviors and activities (e.g., seeking customers) and to respond to similar adaptive tensions, but may also result in inter-dependent goals, such as creating a “business friendly” community or becoming cash flow positive (cf. Cromie, 1987). In addition, the outcomes of some groups of agents will depend on other groups. For instance, an incubator's success is dependent on the success of entrepreneurs who utilize its services (Hochberg, 2016). Further, the innovations that entrepreneurs introduce into an EE can also produce coherence in agents' actions (Muñoz & Encinar, 2014). For instance, an innovation introduced by the experimentation of an entrepreneur, such as a business model innovation, can be accepted and then replicated by other entrepreneurs in an EE, which is a form of “emergent structuration” – that

is, a structure of similarity emerges among agents selecting the same innovation (Miller & Osborn, 2008).

An ecosystem's agents may also share common values (e.g., towards cooperation) or discourse (e.g., shared narratives about the system's past), which also create coherence (Roundy, 2016). Relatedly, the rule sets, or “simple rules” (Bingham & Eisenhardt, 2011), adopted by agents (e.g., “favor cooperation over competition” or “give before taking”; Feld, 2012), which operate as a portfolio of decision making heuristics and are learned by observing the actions of other agents (Axelrod, 1997), can be shared throughout an EE. Even if the rules are not accepted by every agent in the EE, or if there are errors (i.e., failure to send or receive a rule completely), such rules will result in a general tendency to respond in a similar way during interactions, which creates coherence that further structures the ecosystem.

For example, anecdotal evidence suggests that the entrepreneurial community in Boulder, Colorado, is known for its supportive climate for new entrepreneurs; rather than shaming the founders of failed start-ups, these entrepreneurs are welcomed into the community as advisors, mentors, and entrepreneurs-in-residence (Feld, 2012). Support for failed entrepreneurs is a deeply engrained cultural norm of Boulder's start-up community that creates coherence and shapes the general response to new venture failure. Indeed, agents' choice of collaboration over competition in Boulder is one of a series of implicit rules (commonly referred to as the “Boulder Thesis”; cf. Feld, 2012), which has provided the recipe, or configuration, from which the city's EE emerged and coalesced. In general, anything that strengthens the association between agents, such as sharing common intentions, narratives, patterns of thinking, or values, will increase the coherence of an EE and act as a force that creates the complex system, structures it, and acts against disordering tendencies (Roundy et al., 2017).

4.2.2. Injections of resources

As complex adaptive systems, EEs are open and, therefore, can be influenced by forces outside the permeable boundaries of the system (Spigel, 2017). This suggests that the emergence of EEs can be further influenced by injections of resources, which operate as “control parameters” – forces outside a complex system that can “push” the system and its agents into different behaviors and influence coherence (Rickles, Hawe, & Shiell, 2007). For example, an injection of financial capital into a region that is focused on a specific type of technology, such as additive manufacturing, can cause a collection of entrepreneurs to begin pursuing business models all focused on that same technology and, thus, increase the coherence of their actions. Injections of resources can also increase system coherence by stimulating forces *inside* the EE that further influence agent and system behaviors, such as funding the creation of business incubators (internal coalescing forces are typically referred to in complexity science as “order parameters”) (Goldstein, 1999). For instance, injections of resources can result in EE participants pursuing similar opportunities; in doing so, agents will interact with one another. Internal, micro-interactions can generate “simple rules” and system-level values, which, in turn, guide actions, increase the coherence among agents, and give structure to the EE.

4.3. The coalescence of entrepreneurial ecosystems

To summarize, combining the agent-and system-level forces described above, as a complex adaptive system an EE can be conceptualized as emerging as follows. Initially, entrepreneurs and other agents are engaged in uncoordinated “business as usual” within the general regional economy. Entrepreneurs may develop action plans during this stage based on their intentions and in response to adaptive tensions; however, there is little coherence among plans or behaviors across agents. Nevertheless, continued injection of resources (e.g., capital, information) will begin to be met by more coordinated responses, as the entrepreneurs' behaviors and outcomes influence and are adopted by other agents in the system, such as support organizations,

and as system-level values take hold and influence agents' coherence. Over time, a combination of entrepreneurs' actions (e.g., introducing innovations adopted by other agents), system-level characteristics that create coherence among entrepreneurial activities (e.g., guiding values and rule-sets), and continued injections of resources into the nascent system, stimulate further coherence among agents, which increases the emergent structuration of the EE. Thus, the interdependence of agents, their responses to endogenous and exogenous forces operating at the micro- and macro-levels of the complex system, and their recursive relationships produce the set of interactions out of which an EE emerges. Collectively, these arguments suggest the following proposition.

Proposition 7. The emergence of entrepreneurial ecosystems as complex adaptive systems is influenced by three forces: (a) entrepreneurs' intentionality and adaptive tensions, (b) coherence in entrepreneurial activities, and (c) injections of resources.

5. Discussion

Past research has made strides in identifying the components of EEs and exploring the connections among them. However, as scholars adopting the complexity lens have suggested, “by examining entrepreneurial components in isolation from each other, we lose sight of the [...] wholeness of entrepreneurship [and] the isolation of individual parts of the system (in analysis) does not reveal the casual mechanisms in the system” (Anderson et al., 2012). As we have theorized, this statement applies beyond individual- and organization-level entrepreneurship to EEs. Furthermore, our theorizing represents not only a framework for analyzing EEs, but it also addresses the criticisms of entrepreneurship research of being “disjointed,” “fragmented,” and as not having a unified theory that can tie together individual-, venture-, and system-level findings (Fuller & Moran, 2001). By focusing on the concept of emergence, we have sought to tie together complexity-based entrepreneurship research at these levels. Furthermore, the definition we develop also highlights the fact that “entrepreneurial ecosystem” is not a concept – or title – that is only bestowed upon certain cities or regions. Entrepreneurial activities in cities of any size and scope can be dependent on a complex system of inter-related forces.

At the individual-level of analysis, a growing stream of academic and practitioner research examines entrepreneurs' use of experimentation in the development of business models (Kerr, Nanda, & Rhodes-Kropf, 2014; Osterwalder, 2004). However, given its micro-focus, this work has developed in isolation from the research on EEs. Our theory identifies the important connections that exist between entrepreneurial intentions, experimentation, and actions and the emergence of the EEs in which entrepreneurs operate.

5.1. Methods for studying EEs as complex systems

The most critical direction for future research is to empirically verify the ideas put forth in this paper. To facilitate the testing of our theory, we provide an outline of how scholars might conduct studies that use elements of complexity theory to analyze EEs. We follow other scholars who have argued that new theories often require new methodologies (Ketchen Jr, Boyd, & Bergh, 2008). New methods are especially needed when examining complex phenomena because their non-linearity and dynamics suggest that traditional methods, often based on linearity assumptions, are not appropriate (Berger & Kuckertz, 2016b).

After reviewing the complex adaptive systems literature and research at the intersection of complexity and entrepreneurship, we believe that scholars would be well-served by studying EEs using mix-methods approaches (Najmaei, 2016) that leverage both quantitative and qualitative techniques. Specifically, below we outline the use of three methods: qualitative comparative analysis (QCA), agent-based modeling, and interpretivist qualitative research.

5.1.1. Qualitative comparative analysis

Qualitative comparative analysis is a “configurational approach” based on Boolean algebra and used to study phenomena that are “best understood as clusters of interconnected structures and practices, rather than as modular or loosely coupled entities whose components can be understood in isolation” (Fiss, 2007: 1180; Ragin, 1987). Although research has tended to treat EEs as being composed of isolated or only semi-connected components (cf. Motoyama & Knowlton, 2017 for an exception) and as characterized by singular causation or linearity, EEs emerge from nonlinear and dynamic combinations of sets of variables. Thus, different configurations of ecosystem components are likely to lead to “causal asymmetry” and “to equifinal outcomes, which are not necessarily the same configurations explaining non-outcomes[s]” (Berger, 2016: 2; Ragin, 2008). This suggests that traditional methods of analysis, based on assumptions of linearity, are ill-suited for studying EEs because of multicollinearity, non-normality of data, and an inability to account for contrarian cases, all which can generate misleading results (Olya & Mehran, 2017; Woodside, 2013). Although most QCA studies in management and organization studies have focused on the organization-level (Berger, 2016), the method is also applicable to ecosystems.

To analyze EEs using QCA would involve roughly the following steps. First, researchers would identify a configuration-driven research question. For example, QCA could potentially help scholars examine questions such as, “under what conditions will an EE produce high and low levels of new venture creation activity?” or “under what conditions will an EE be resilient to disturbances?” A sample of EEs could then be assembled. One of the advantages of QCA is that an intermediate number of cases (e.g., 20–50) can be used. When predicting a dependent variable, such as EE resilience or new venture activity, conventional methods (e.g., regression analysis) often implicitly assume that the conditions predicting high scores are mirror opposites of the forces responsible for low scores (Olya & Mehran, 2017). However, QCA does not rely on such symmetries. Instead, researchers seek to identify several conditions (five to six conditions are recommended; Berger, 2016) to predict the outcome. For example, factors isolated in prior research (e.g., such as the presence of venture capital, large firms, and research universities; Neck et al., 2004) could serve as focal conditions. The next step is to calibrate measures of these characteristics by transforming measures of the concepts and assigning membership as part of a condition (Berger, 2016). After transforming the values of antecedents into set membership, a truth table is generated, which provides a list of combinations of the conditions that lead to the focal outcomes (Ragin, 2008). These results could reveal, for instance, that there are five sufficient and consistent “recipes” for EEs with high new venture activity. In contrast, other configurations of characteristics might exist which, on the surface similar, are associated with low new venture activity.

5.1.2. Agent-based computational modeling and optimization

Scholars from a variety of disciplines have used formal simulation models to explore the behavior of complex systems. A popular subset of these techniques is adaptive agent models, which assume that agents exhibit stochastic nonlinear behaviors and change over time through adaptive improvements (cf. McKelvey, 1999: 7). These types of models have begun to receive attention in management research (e.g., work on NK modeling and “rugged” learning landscapes; Kauffman & Johnsen, 1991) and, to a lesser extent, in studies of individual-level entrepreneurship characteristics, such as alertness (e.g., spin-glass models; Minniti, 2004).

With adaptive agent modeling, agents' attributes (e.g., financial resources) are typically assigned by random draw, and then agents can conduct “searches” over a simulated number of rounds in which they interact with other agents who are subject to processes also determined at random (McKelvey, 1999). System “outcomes are the result of individual agent attributes, search results, larger system attributes, and environmental constraints” (McKelvey, 2004: 328). In constructing

adaptive agent models, a key decision-point that aligns with our theorizing about EEs is the importance of identifying the simple rules that govern agents' actions, which are generally grounded in existing empirical research (McKelvey, 2004). In addition to rules, the types of agents, their search space, the agents' objectives, and their level of connectivity all help to determine the coherence (or disorder) that emerges in the system (McKelvey, 2004).

The critical steps in using such methods to model the behaviors of EEs involve identifying the key components of the ecosystem (prior EE work could be leveraged for this step), specifying the relationships between components by distilling the interactions to their critical features, and then observing how the model of the EE changes over many simulated iterations when the starting conditions and relationship parameters are altered. It is important to note that, as theorized, EEs emerge from nonlinear processes and interactions. One temptation when attempting to construct adaptive agent models is to try to (fully) linearize the relationships in the system, thus reducing or removing the system's complexity. Unfortunately, doing so also removes the ability to generate accurate insights into the nuanced behavior of ecosystems.

5.1.3. Qualitative and interpretivist methods

QCA and agent-based modeling can be paired with other methods; indeed, scholars using both methods have called for studies to use mixed-methods (e.g., Davis, Eisenhardt, & Bingham, 2007). Although quantitative methods have been the dominant approach to studying CAS in other disciplines, qualitative methods (and data) are also well-suited for the study of EEs as complex systems, for several reasons. First, as we theorize, EEs are characterized by nonlinear dynamics, feedback loops, and complex, multi-level interactions. The flexibility of qualitative methods (e.g., semi-structured interviewing, ethnographic observation) makes them particularly appropriate for studying such phenomena. Specifically, the richness of qualitative data can enable scholars to unpack the multi-faceted, temporally unfolding characteristics of EEs and tease apart the causal relationships in such ecosystems (cf. Graebner, Martin, & Roundy, 2012). Second, communication among agents is a process that both creates and constitutes the interactions of an EE. It stands to reason that it is through narratives, discourse, and other linguistic constructs that the knowledge, values, and culture that form the bedrock of an EE are passed between participants. As other studies of entrepreneurial phenomena have demonstrated (e.g., Martens, Jennings, & Jennings, 2007) qualitative data are particularly effective in capturing such discourse. Finally, the vividness and concreteness of qualitative data make it well suited for paring with some of the more abstract quantitative methods, such as agent-based modeling and QCA, since qualitative data can illustrate abstract ideas or make conceptual frameworks more concrete and understandable (Graebner, Martin, & Roundy, 2012).

5.2. Implications for policymakers and practitioners

Creating effective policies for EEs requires an integrative view that acknowledges their complexity. Although complexity research is largely based on contexts and theories that, on their surface, may seem abstract and removed from the daily decisions and activities of entrepreneurs, a complex systems-based conceptualization of EEs has several concrete implications for those seeking to grow an EE and for entrepreneurs.

First, acknowledging that EEs are complex adaptive systems suggests that policymakers should be mindful of the type of metrics used in their evaluations of such systems. Work on other complex systems has found that they cannot be effectively assessed using simple, “count-based” metrics. For instance, Levin (1998) describes this challenge in the context of ecological ecosystems and explains that when attempting to measure a system's diversity, “simple species counts [...] do not alone capture the features that are most important for sustaining ecosystem functioning [as] [n]ot all species are equally important to the

maintenance of key processes” (429). We would argue that this statement is equally applicable to EEs. Traditional metrics used to measure economic activity, such as the number of jobs created or counts of new venture foundings, may not fully capture the functioning of the system. In the same way that the behavior of complex systems cannot be captured with simplistic equations, it stands to reason that the functioning of an EE requires metrics that assess the characteristics of the ecosystem *directly related to its complexity*. As discussed, the behaviors of EEs are largely a result of the complexity of the interactions among system components. Thus, to measure an EE's ability to produce certain behaviors may require metrics that capture the depth and breadth of those interactions, such as metrics that measure features of the network of connections in the EE, or the extent to which entrepreneurs are encouraged to experiment. Thus, it may be necessary to look past simple, surface-level characteristics of the system to measure its deeper-level properties.

Most current metrics (e.g., the number of ventures created) seem focused on measuring one aspect of the growth of an EE. However, as in other systems, growth is dependent on maintaining an underlying homeostasis. Using the human body (also a complex adaptive system) as an example, humans are able to live and grow only if their bodies maintain a stable environment as indicated by various vital signs; for instance, if a body cannot maintain its temperature (a homeostasis metric) within a narrow range of 98.6 degrees, changes in weight (a growth metric) do not matter because life cannot exist. But what are the “vital signs” relevant to an emerging EE? Underneath the growth of an EE, there may exist homeostasis-oriented factors, which are just as critical as growth factors to the emergence of the system. Indeed, as with body temperature, relatively small changes in homeostatic conditions may have immense consequences for EE emergence, which suggests the importance of examining the factors that *sustain* an EE. For example, the quality of a region's infrastructure – i.e., public transportation, education system – may play an essential role in supporting its EE.

The adaptability of EEs as complex systems can have concrete consequences for entrepreneurs. Indeed, the situation entrepreneurs face in developing new ventures in EEs is akin to playing billiards and the table surface being altered while the ball is rolling. The challenges associated with operating in a constantly adapting system highlight the limitations of overly rigid, inflexible, formal planning and the advantage of approaches that emphasize flexible startup development processes. Our theorizing also emphasizes the importance of promoting certain individual-level activities and macro-level system properties. Specifically, it suggests that at the micro-level of an ecosystem, entrepreneurs should be encouraged to employ experimentation-based venture development methods because doing so can introduce innovations into the system. Similarly, system-level cultural rules such as “help other EE participants” or “give before taking” should be encouraged because they can increase ecosystem coherence.

In sum, our theorizing represents a first-cut at conceptualizing entrepreneurial ecosystems as complex adaptive systems. By introducing the lens of complexity to the study of such systems, we hope to have opened new avenues of research on entrepreneurial ecosystems, which deepen our understanding of a phenomenon that is central to the health of modern economies.

Declaration of interests

None.

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